Tran The Duong

Computer Vision | Multi-Object Tracking | Robotics and Control
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M.S. student at KAIST developing Transformer-based multi-object tracking systems, with a background in robot dynamics, optimal control, and reinforcement learning.

Education

KAIST
M.S. in Electrical Engineering | *Daejeon, South Korea* Sep. 2024 – Present
Expected graduation: Feb. 2027. Advisor: Prof. Dong Eui Chang.

Hanoi University of Science and Technology
B.S. in Automatic Control | *Hanoi, Vietnam* Sep. 2019 – May 2024
GPA: **3.89/4.00**. Thesis advisor: Assoc. Prof. Dao Phuong Nam.

Technical Skills

Programming: C, C++, Python | **Deep learning:** PyTorch.
Modeling and simulation: MATLAB, Simulink.
Robotics: Robot dynamics, control, and state estimation.
Reinforcement learning: Strong foundational knowledge.
Languages: Vietnamese (native), English.

Research Experience

Graduate Researcher, Control Laboratory, KAIST
Multi-object tracking; robotics and control Sep. 2024 – Present
Tracking: Oct. 2025 – Present. Robotics and control: Sep. 2024 – Oct. 2025.

Master's thesis (ongoing): End-to-End Transformer-Based Multi-Object Tracking for Drone and Bird Surveillance

- Developed **BeliefCO**, combining learned motion uncertainty and object existence with Transformer-based tracking and probabilistic data association.
- Built WOSDETC tracking annotations and recording-level splits; evaluated across four aerial-tracking benchmarks with controlled detector settings and component ablations.
- Achieved test HOTA scores of **72.32 on BFT**, **62.91 on WOSDETC**, and **65.77 on BSB**.

Bayesian Motion Prompting *Python, PyTorch*

- Extended MOTIP with motion uncertainty and motion-based identity association; built training and evaluation pipelines for DanceTrack, SportsMOT, and bird flock tracking.

Manuscripts in Preparation

Three manuscripts being prepared for planned submission to CVPR 2027.

DUET: Overlap-Aware Likelihood-Ratio Association for Online Multi-Object Tracking
Maintaining object identities during overlaps.

Lifespan: Memory-Guided Association and Track Management for Online Multi-Object Tracking
Preserving identity continuity when objects disappear and return.

FarID: Learning Identity Beyond Spatial Proximity for Online Multi-Object Tracking
Learning persistent target identity across long occlusions.

Robotics and Control Projects

Wheeled-Bipedal Robot Modeling and CLF-MPC Control
MATLAB/Simulink, CasADi, C++, Gazebo Sep. 2024 – Oct. 2025

- Derived nonlinear wheeled inverted pendulum dynamics and implemented constrained model predictive control with Control Lyapunov Function conditions, actuator limits, and smoothness penalties.
- Built a Gazebo–Simulink interface using C++ and Protocol Buffers to exchange joint states, IMU measurements, and torque commands; generated polynomial height trajectories.

RS-JiLQG: Risk-Sensitive Control with Slip Jumps
Graduate stochastic control project | Python, NumPy Sep. 2024 – Oct. 2025

- Co-developed a risk-sensitive jump iterative Linear Quadratic Gaussian formulation for nonlinear systems with diffusion noise and intermittent slip disturbances.
- Implemented a solver with backward policy updates, regularization, and line search; evaluated closed-loop Monte Carlo rollouts against LQR baselines with Kalman-filter state estimation.

PX4 Offboard Control through ROS 2
Python, rclpy, PX4 message interfaces Nov. 2024

- Implemented offboard heartbeats, vehicle commands, and position setpoints, with arming, takeoff, and landing logic plus vehicle-state and command-acknowledgment monitoring.

Earlier Engineering and Research Experience

Engineering Intern, Viettel Aerospace Institute
Embedded systems software Sep. 2023 – Feb. 2024

- Designed user interfaces for embedded systems; worked with C++, MATLAB, and Simulink.

Undergraduate Researcher, HUST
Reinforcement learning and control 2022 – 2024

- Studied output-feedback adaptive dynamic programming (ADP) and robust integral-of-sign-error reinforcement learning control for a two-wheeled inverted pendulum in the undergraduate thesis.
- Investigated ADP-based control for wheeled robots with wheel slip and wind turbines with disturbance observers; designed a PID controller for a resistance furnace.

Additional Programming Projects

C++ Programming Exercises
C++, Git Apr. 2025
Maintained introductory C++ examples for language practice and version control.

Pattern Recognition with RBM Features and Neural Classification
Python, NumPy, TensorFlow | KAIST coursework Dec. 2024
Implemented restricted Boltzmann machine feature learning and a neural classifier for MNIST, including sampling, normalization, and train/validation/test splits.

Awards and Honors

- **Best Graduation Thesis**, HUST thesis defense council, Feb. 2024.
- **Academic excellence scholarships**, HUST, six semesters (2020–2023); corporate scholarship for outstanding students (2023).